PET-MR synergy

1M = 10^{-3} M

Date	2 Nov. 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	IQ			
Given Radioactivity	27.05 mCi 26.25	9:00		
Dose (full 59g)		Time	mCi	
	Full syringe dose	9:08 10	122 μ 118 μ	hot cold
	Empty syringe	9:15	21 μ 20 μ	
Acquisition Time	20 min (PET) / 7 min (MR)			
Reconstruction Parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.25 mm	18	—	

Date	2 Nov. 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	~~Rat~~ Rat-sized Perchro			
Given Radioactivity	26.25	9:00		
Dose (Full 159g)		Time	mCi	
	Full syringe dose	9:44	175 μCi	
	Empty syringe	9:44	20 μCi	Total = 95 mj = 96
Acquisition Time	20 min (PET) / 7 min (MR)			
Reconstruction Parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.25 mm	18	—	

* SC (sss) + AC afterward

Date	2 Nov. 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	IQ phantom			
Given radioactivity	26.25 mCi 9:00			
Dose 50% 59g		Time	mCi	
	Full syringe dose	11:09	decay	
	Empty syringe		total 92 mCi	
Acquisition Time	10 min PET, 3 mm MR			
Reconstruction parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.25 mm	18	—	—

Date	2 Nov. 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	Rat-Sized Derenzo			
Given radioactivity	26.25 mCi 9:00			
Dose 50% 157g		Time	mCi	
	Full syringe dose	11:25	decay	
	Empty syringe		total: 51 mCi	
Acquisition Time	10min PET. 3min MR			
Reconstruction parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.25 mm	18	—	—

SC (sss) + AC afterward

Date	2 Nov. 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	IQ phantom			
Given Radioactivity	26.75 mCi 9:00			
Dose ~10% 60g		Time	mCi	
	Full syringe dose	suck + decay 11:46	total 31 mCi	
	Empty syringe dose			
Acquisition Time	PET 2 min MR 3 min			
Reconstruction Parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.75 mm	18	—	—

Date	2 Nov. 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	Rat-size L Perem80			
Given Radioactivity	26.75 mCi 9:00			
Dose 100% 158g		Time	mCi	
	Full syringe dose	suck + decay 11:59	total 12 mCi	
	Empty syringe dose			
Acquisition Time	PET 2 min MR 3 min			
Reconstruction Parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.75 mm	18	—	—

SC (sss) + AC afterward

SC (sss) / w.o. SC (sss) (total 4 pieces of data)
AC on for all (post-processing)

Date	Nov 2, 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	Three cold rod Phantom			
Given Radioactivity	16.25 mCi 9:00			
Full Dose 55g		Time	mCi	
	Full syringe dose	10:19	117 M	
	Empty syringe dose	10:19	17 M	
Acquisition Time	PET only (20min)			
Reconstruction Parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.25 mm	18	—	

Date	Nov. 2 2026			
Machine Name	Bruker 7T PET/MR			
Type of Phantom	Concentric circle phatom			
Given Radioactivity	16.25 mCi 9:00			
Full Dose 194g		Time	mCi	
	Full syringe dose	10:38	239 M	
	Empty syringe dose	10:38	20M	
Acquisition Time	PET only 20min			
Reconstruction Parameters	Reconstruction method	Iteration number	Subsets	Filter
	MLEM 0.25 mm	18	—	